

The Least Atom Determines the Optimal Hypercontractive Constant

Automated Math Discovery Candidate Proof

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Status. Candidate full solution, likely valid pending expert review. The new step is the strict monotonicity in the Bernoulli bias of the exact constant obtained in the source paper.

Source and Conjecture

This packet concerns the unnumbered conjecture on page 8 of Cao et al. [1], arXiv:2602.17248v2. For a probability measure μ on a finite set, put

$$\sigma_{p,q}(\mu) = \max \left\{ 0 < r < 1 : \|\mathbb{E}_\mu + r(I - \mathbb{E}_\mu)\|_{L^p(\mu) \rightarrow L^q(\mu)} \leq 1 \right\}, \quad 1 < p < q < \infty.$$

If $\lambda \in (0, 1/2]$, write $\sigma_{p,q}(\lambda)$ for the same constant on the two-point space with masses λ and $1 - \lambda$. The conjecture states that if the least atom of μ has mass λ , then

$$\sigma_{p,q}(\mu) = \sigma_{p,q}(\lambda).$$

Let μ be a probability measure on a finite set Ω , for any $1 < p < q < \infty$, define

$$\begin{aligned}\sigma_{p,q}(\mu) &:= \max \{0 < r < 1 : \|\mathbb{E}_\mu + r(Id - \mathbb{E}_\mu)\|_{L^p(\mu) \rightarrow L^q(\mu)} \leq 1\} \\ &= \inf \{\sigma_{p,q}(X) : X \text{ is random variable defined on } \Omega \text{ with } \mathbb{E}_\mu(X) = 0\}.\end{aligned}$$

Since $\sigma_{p,p^*}(\lambda)$ given in (1.11) is increasing in $\lambda \in (0, 1/2)$, using [Wol07, Theorem 3.1], we have

Corollary 1.7. *Let μ be a probability measure on a finite set Ω with the mass of the least atom equal to $\lambda \in (0, 1/2)$. Then for all $1 < p < 2$, we have $\sigma_{p,p^*}(\mu) = \sigma_{p,p^*}(\lambda)$ which is given by (1.11).*

Remark. Since $\sigma_{p,q}(\lambda) = \sigma_{p,q}(1 - \lambda)$, Theorem 1.6 gives also $\sigma_{p,q}(\lambda)$ for all $\lambda \in (0, 1) \setminus \{1/2\}$.

Remark. Numerical simulation indicates that $\sigma_{p,q}(\lambda)$ is increasing in $\lambda \in (0, 1/2]$.

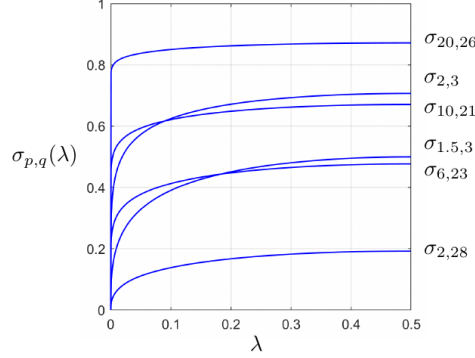


FIGURE 3. Numerical simulation of $\sigma_{p,q}(\lambda)$

Inspired by the simulations in Figure 3, using [Wol07, Theorem 3.1], we formulate the following

Conjecture. Let μ be a probability measure on a finite set Ω with the mass of the least atom equal to $\lambda \in (0, 1/2)$. Then for all $1 < p < q < \infty$, we have $\sigma_{p,q}(\mu) = \sigma_{p,q}(\lambda)$.

1.3. Noise operators. The noise operator plays an important role in λ -biased Fourier analysis, and provides another key example of hypercontractive random variables. For $\lambda \in (0, 1/2]$, let $\Omega = \{-1, 1\}$ and let π_λ be the distribution on Ω defined by

Proof Intuition

Wolff's two-valued-extremizer theorem already sandwiches the finite-space constant between

$$\inf_{\alpha \in [\lambda, 1/2]} \sigma_{p,q}(\alpha) \quad \text{and} \quad \sigma_{p,q}(\lambda).$$

Thus the conjecture follows once the biased two-point constant is known to be increasing in the smaller mass. This monotonicity was supported numerically in the source paper but was left open.

The exact two-point formula in the source paper permits an envelope derivative. If $a = \lambda/(1 - \lambda)$ and (x, y) is a critical extremizer, then a direct differentiation collapses to

$$\frac{d}{da} \log \sigma_{p,q}(\lambda) = \frac{K_p(x) - K_q(y)}{a}, \quad K_s(t) = \frac{1 - st^{s-1} + (s-1)t^s}{s(1-t)(1-t^{s-1})}.$$

The apparent two-parameter sign problem becomes transparent after writing $t = e^{-A}$:

$$K_s(e^{-A}) = \frac{1}{2} + \frac{h(A) - h((s-1)A)}{sA}, \quad h(u) = \frac{u}{2} \coth \frac{u}{2}.$$

The function h is increasing and convex. The second critical equation forces the two logarithmic coordinate pairs to cross: the first coordinate increases and the second decreases. Monotonicity

and convexity of h then give $K_p(x) > K_q(y)$. Duality handles exponents below 2, while exponents straddling 2 have opposite signs around $K = 1/2$.

Main Result

Theorem 1. *Let $1 < p < q < \infty$.*

1. *The function $\lambda \mapsto \sigma_{p,q}(\lambda)$ is strictly increasing on $(0, 1/2]$.*
2. *If μ is a probability measure on a finite set and its least atom has mass $\lambda \in (0, 1/2)$, then*

$$\sigma_{p,q}(\mu) = \sigma_{p,q}(\lambda).$$

Consequently, the conjecture quoted above has an affirmative answer.

The Envelope Derivative

Set $a = \lambda/(1 - \lambda) \in (0, 1)$. For $s > 1$ and $0 < t < 1$, define

$$\begin{aligned} F_s(a, t) &= \frac{1}{s} \log(a + t^s) - \log(a + t) + \left(1 - \frac{1}{s}\right) \log(1 + a), \\ B_s(a, t) &= \frac{(1 - t)(1 - t^{s-1})}{a + t^s}, \\ K_s(t) &= \frac{1 - st^{s-1} + (s - 1)t^s}{s(1 - t)(1 - t^{s-1})}. \end{aligned}$$

Theorem 1.6 and the minimum formulation following it in [1] say that an active nonconstant two-point extremizer is described by $x, y \in (0, 1)$ satisfying

$$F_p(a, x) = F_q(a, y), \quad B_p(a, x) = B_q(a, y), \tag{1}$$

and its critical constant is

$$r = \frac{(1 - y)(a + x)}{(1 - x)(a + y)}. \tag{2}$$

Lemma 1 (Derivative identity). *Along every smooth active branch of solutions to (1),*

$$\frac{d}{da} \log r = \frac{K_p(x) - K_q(y)}{a}.$$

The same formula is the envelope derivative at any differentiability point of the optimal value.

Proof. For fixed (a, x) , the first equation in (1) determines y locally because

$$\partial_t F_s(a, t) = -\frac{aB_s(a, t)}{(1 - t)(a + t)} < 0. \tag{3}$$

Let $R(a, x) = \log r$, with $y = y(a, x)$ determined by that first equation. A direct use of (3) gives

$$\partial_x R = \frac{a + 1}{(1 - x)(a + x)} \left(1 - \frac{B_p(a, x)}{B_q(a, y)}\right).$$

Thus the second equation in (1) is precisely the stationary condition $\partial_x R = 0$.

Write $B = B_p(a, x) = B_q(a, y)$. Differentiating in a while holding x fixed and using the first equation gives

$$\partial_a R = \frac{1}{a+x} - \frac{1}{a+y} + \frac{a+1}{aB} (\partial_a F_p(a, x) - \partial_a F_q(a, y)). \quad (4)$$

The elementary identity

$$\partial_a F_s(a, t) + \frac{aB_s(a, t)}{(a+1)(a+t)} = \frac{B_s(a, t)K_s(t)}{a+1} \quad (5)$$

reduces (4) to

$$\partial_a R = \frac{K_p(x) - K_q(y)}{a}.$$

Along a critical branch the $x'(a)$ contribution vanishes because $\partial_x R = 0$, proving the assertion. The envelope statement is the standard finite-dimensional envelope theorem applied to the one-variable two-point variational problem. $\square$

The Sign of the Derivative

Lemma 2. *The function*

$$h(u) = \frac{u}{2} \coth \frac{u}{2}, \quad u > 0,$$

is strictly increasing and strictly convex.

Proof. Differentiation yields

$$h'(u) = \frac{1}{2} \coth \frac{u}{2} - \frac{u}{4} \operatorname{csch}^2 \frac{u}{2} > 0,$$

where positivity is equivalent to $\sinh u > u$. A second differentiation gives

$$h''(u) = \frac{u \cosh(u/2) - 2 \sinh(u/2)}{4 \sinh^3(u/2)} > 0.$$

The numerator in the last expression vanishes at 0 and has derivative $(u/2) \sinh(u/2) > 0$. $\square$

For $A, B > 0$, put

$$\mathcal{K}(A, B) = \frac{1}{2} + \frac{h(A) - h(B)}{A + B}.$$

On the region $0 < A \leq B$, Lemma 2 implies

$$\begin{aligned} \partial_A \mathcal{K}(A, B) &= \frac{(A+B)h'(A) + h(B) - h(A)}{(A+B)^2} > 0, \\ \partial_B \mathcal{K}(A, B) &= \frac{h(B) - h(A) - (A+B)h'(B)}{(A+B)^2} < 0. \end{aligned}$$

For the second inequality, convexity gives $h(B) - h(A) \leq (B-A)h'(B) < (A+B)h'(B)$.

Proposition 1. *Suppose $2 \leq p < q < \infty$ and (x, y) is an active critical pair in (1). Then $K_p(x) > K_q(y)$. The same conclusion holds when $1 < p < 2 < q$.*

Proof. First note from (2) that $0 < r < 1$ is equivalent to $x < y$.

Assume first that $2 \leq p < q$. Set

$$A = -\log x, \quad B = (p-1)A, \quad C = -\log y, \quad D = (q-1)C.$$

Then $A > C$, $B \geq A$, and $D \geq C$. The second equation in (1) becomes

$$\frac{(1 - e^{-A})(1 - e^{-B})}{a + e^{-(A+B)}} = \frac{(1 - e^{-C})(1 - e^{-D})}{a + e^{-(C+D)}}. \quad (6)$$

If $B \geq D$, then the numerator on the left of (6) is strictly larger than the numerator on the right, while its denominator is strictly smaller. This is impossible. Hence

$$A > C, \quad B < D.$$

An elementary rearrangement gives

$$K_s(e^{-u}) = \frac{1}{2} + \frac{h(u) - h((s-1)u)}{su}. \quad (7)$$

Consequently $K_p(x) = \mathcal{K}(A, B)$ and $K_q(y) = \mathcal{K}(C, D)$. Starting at (C, D) , decreasing the second coordinate from D to B increases $\mathcal{K}$, and then increasing the first coordinate from C to A increases it again. Therefore $K_p(x) > K_q(y)$.

Now suppose $1 < p < 2 < q$. In (7), $(p-1)A < A$, so the strict increase of h gives $K_p(x) > 1/2$. Similarly $(q-1)C > C$, so $K_q(y) < 1/2$. The conclusion follows. $\square$

Proposition 2 (Strict bias monotonicity). *For every $1 < p < q < \infty$, the function $\lambda \mapsto \sigma_{p,q}(\lambda)$ is strictly increasing on $(0, 1/2]$.*

Proof. For $2 \leq p < q$, Lemma 1 and Proposition 1 show that every active envelope derivative is strictly positive. The same holds directly for $p < 2 < q$.

For completeness, the exact constant is a locally Lipschitz value function of a on compact subintervals of $(0, 1)$: it is the minimum of the smooth one-variable two-point crossing values, and the active minimizers supplied by [1, Theorem 1.6] lie in the interior. Thus the envelope theorem applies almost everywhere. A locally Lipschitz function cannot decrease or be constant on a nondegenerate interval when all its almost-everywhere envelope derivatives are strictly positive. Hence the increase is strict.

If $1 < p < q \leq 2$, self-adjointness of the two-point noise operator gives

$$\sigma_{p,q}(\lambda) = \sigma_{q',p'}(\lambda), \quad q' = \frac{q}{q-1}, \quad p' = \frac{p}{p-1}.$$

Here $2 \leq q' < p'$, so the already proved case applies. Finally, $a = \lambda/(1-\lambda)$ is strictly increasing in λ . $\square$

Completion of the Finite-Space Conjecture

In the notation $1 < p < q$ used here, Wolff's Theorem 3.1 [2] states that if the least atom of a finite probability measure μ has mass λ , then

$$\inf_{\alpha \in [\lambda, 1/2]} \sigma_{p,q}(\alpha) \leq \sigma_{p,q}(\mu) \leq \sigma_{p,q}(\lambda). \quad (8)$$

(Wolff writes the target exponent first, so his displayed indices are in the reverse order.) By Proposition 2, the infimum on the left of (8) equals $\sigma_{p,q}(\lambda)$. Both inequalities are therefore equalities, proving Theorem 1.

Verification Notes

The accompanying script `code/verify_derivative.py` checks the two algebraic identities (5) and (7) symbolically. It also performs randomized finite-precision checks of the coordinate-crossing argument and compares the analytic derivative sign with finite differences of independently optimized two-point constants. These computations are guards against transcription or sign errors; they are not used as proof.

The most important points for expert review are:

1. the passage from the exact variational formula of [1] to the locally Lipschitz envelope statement in Proposition 2;
2. the translation of Wolff’s exponent order in (8);
3. the strict coordinate crossing $A > C$ and $B < D$ obtained from (6).

Novelty Check and Limitations

On July 21, 2026, bounded web searches were run for the exact phrases and close variants “ $\sigma_{p,q}(\lambda)$ monotonicity biased Bernoulli hypercontractivity”, “optimal hypercontractive constants least atom conjecture”, the title of the source paper, and Wolff’s title together with “monotonicity”. The searches found the source paper, Wolff’s 2007 paper, and general background references, but no separate paper proving the stated monotonicity or answering the conjecture. The local run indexes likewise had no prior packet for arXiv:2602.17248 or this claim. Novelty confidence is therefore moderate rather than definitive, and the result remains a candidate until expert review.

The proof relies on the exact two-point variational formula in [1] and on Wolff’s Theorem 3.1. It does not reprove those two inputs.

References

- [1] J. Cao, S. Fan, Y. Han, Y. Qiu, and Z. Wang, *The optimal hypercontractive constants for $\mathbb{Z}_3$ and biased Bernoulli random variables*, arXiv:2602.17248v2, 2026.
- [2] P. Wolff, *Hypercontractivity of simple random variables*, *Studia Math.* **180** (2007), no. 3, 219–236, doi:10.4064/sm180-3-3.